PhD Notes and Pre-reading

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Document Information and Key

This document is shared with Malcolm Joyce (project supervisor), Fleur Hughes (programme manager), James Lewis (programme coordinator), Dan Doherty (taught programme lead) and Alyanna Dela Cruz (PLANET PGR at Lancaster).

It is not a formal document for submission and the content is not necessarily original. For example, the majority of chapter 1: *Nuclear Data* can be taken to be skeleton notes from Jenkins' and Wood's *Nuclear Data* series. It is to be used as a personal reference tool, diary and as proof of engagement. As such, formal references will not be given, only in the case of copied or adapted diagrams and where it is useful for self-reference. Instead of references, an informal list of sources will be given in the introduction to each chapter.

AI will not be used for generating or proof-reading text, writing any LaTeX markup or for formatting. For time's sake, it will be used for reproducing editable diagrams where I would like to adapt them in some way. For example, figure 1.1 was reproduced by using Claude Sonnet 5.5. Where this is done, a ✻ symbol will be shown (closest dingbat to Anthropic's logo). I have chosen to work with Claude as it is familiar. I have searched for Lancaster's recommended AI tool, but cannot find any definitive answer. In 2023, Lancaster's digital digest included mention of an online training course, 12 Days of AI, Claude is recommended on Day 4.

Worked Example ●○○

Solutions to assigned exercises. The 'difficulty dots' match the asterisks given in *Primer*. For multiple parts of varying difficulties, the highest number is taken.

Derivation ●●○

Derivations of equations.

Historical Context

Historical context and background information.

Part I

Taught Programme: Baseline Training

1 Nuclear Data

1.1 Introduction

The following notes are based on *Nuclear Data: A Primer* by Jenkins and Wood in preparation for the PLANET baseline training lectures in nuclear structure. Including additions from Krane, *Nuclear Physics in a Nutshell* by Bertulani, chapters 3, 9 and 11 of *A Modern Primer in Particle and Nuclear Physics* by Terranova, third year Nuclear II lecture notes (Andreyev) and fourth year Quantum V lecture notes (Wilson-Rae/Cavill).

1.2 Nuclear Data Sets

Nuclear data are mostly organised by mass chains or A chains i.e. data is evaluated for nuclear isobars (sometimes an A chain will contain a single isotope). Recall that there are various categories of isotopes - groups of nuclei that share a common property:

- Isobar: Same mass, $A = N + Z$.
- Isotope: Same proton number, Z.
- Isotone: Same neutron number, N.
- Isomer: Same mass and proton number, but with some nucleons in an excited state e.g. metastable Technetium-99, ^{99m}Tc.

Instead of A chains, groups occassionally produce 'horizontal' compilations and evaluations i.e. a compilation or evaluation of a specific nuclear property across an entire range of *A*, *Z* and *N* - referred to as a mass surface. An example of a horizontal compilation is that of nuclear masses, maintained by the Atomic Mass Evaluation (AME) network. Examples of commonly used nuclear databases:

- Evaluated Nuclear Structure Data File (ENSDF) and Experimental Unevaluated Nuclear Data List (XUNDL) maintained by the National Nuclear

Data Centre (NNDC). It also publishes evaluations by A chain in the journal *Nuclear Data Sheets* and has an interactive plotting tool, NuDat3.

- Specialised evaluations appear in journals such as *Atomic Data and Nuclear Tables* and *Chinese Physics C*. Review articles on special topics appear in *Physics Reports*, *Progress in Particle and Nuclear Physics*, *Reports on Progress in Physics* and *Reviews of Modern Physics*.

1.3 Gross Properties of Nuclei

Gross properties are those which involve the entire nucleus - radius and mass fall within this definition.

1.3.1 Measuring Radii

A nucleus doesn't have a sharp surface - it is diffuse. This is deduced from inferring charge density distributions, $\rho(r)$, using elastic electron scattering. Charge density as a function of the distance from the centre of the nucleus obeys a typical Fermi distribution.

$$\rho(r) = \frac{\rho_0}{1 + e^{r - \frac{R_{1/2}}{a}}}$$

Where $\rho(R_{1/2}) = \frac{1}{2}\rho_0$ and a gives the width of the surface diffuseness. The width of the interval where density decreases from 90% of ρ_0 to 10% of ρ_0 is $4.4a$. These diffuse effects are mainly due to p-n interactions. Nucleon density is approximately constant inside the nucleus. For most nuclei, there are more neutrons than protons so the individual density distributions would both follow the same shape of a Fermi distribution, but the proton curve would be lower.

Instead of electron scattering, x-ray spectroscopy be used on muonic atoms to provide further information on the density distribution. With their increased mass, muons exist closer to the nucleus than electrons so a higher fidelity is possible.

RMS Charge Radius and Isotope Shifts

Since nuclear surface diffuseness is not a fixed geometric property, it is convenient to define an rms charge radius, $\langle r^2 \rangle^{1/2}$, for a given nuclear species (a given Z and N). This is a well-defined quantity and is unique for each nuclear species. These data are usually plotted as isotope shifts - small variations in charge radius along isotopic chains (measured using optical hyperfine spectroscopy), between a given nucleus and a specified isotope of that nucleus i.e. they are plotted for a fixed Z as a function of N.

When $\langle r^2 \rangle^{1/2}$ values are plotted as a function of A, they follow the liquid drop model closely. Following:

$$\left[\frac{5}{3}\langle r^2 \rangle\right]^{1/2} = R_0 A^{1/3}$$

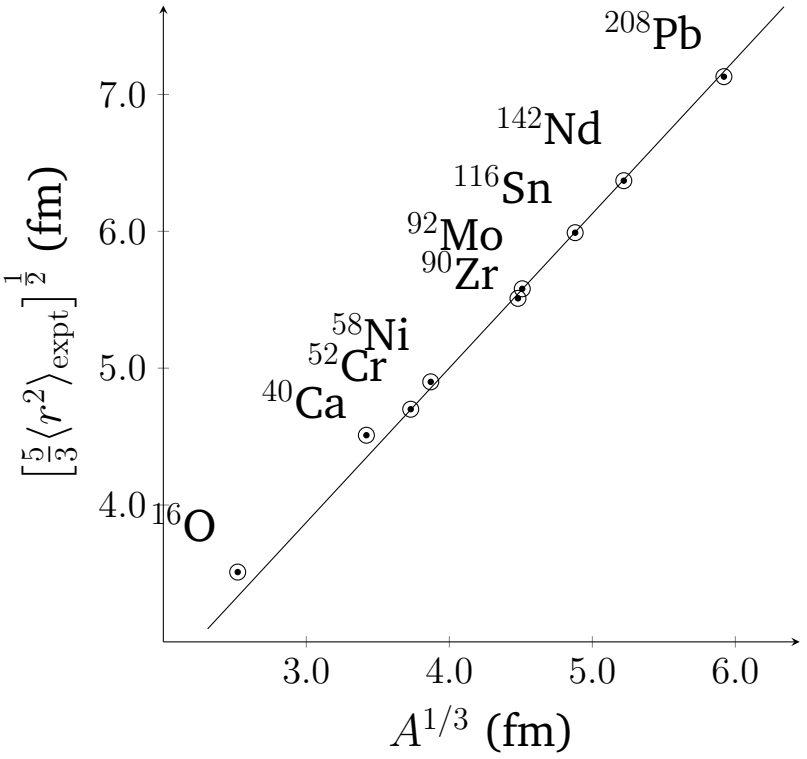


Figure 1.1: Nuclear rms charge radii compared to the liquid drop model. Almost exact for $A \geq 90$, deviation for $A < 90$ due to surface diffuseness. Adapted from fig. 1.7 in *Primer* ❋

Figure 1.1 shows that nuclear size scales with A. If the nuclear binding force were long ranged and attractive, the size of a nucleus would shrink with a larger number of nucleons due to the increased binding. This is what happens for atoms, due to the long ranged Coulomb attraction between the $+Ze$ charge of the nucleus and the electrons.

1.3.2 Measuring Masses

Masses are usually measured for atoms, whereas radii are measured for nuclei. Atoms are easier to handle than nuclei, producing a bare nucleus would involve stripping away all electrons leaving a highly charged ion which would be neutralised by capturing electrons from its environment. Instead, singly charged ions are used because they are easy to maintain and manipulate under a vacuum with E and B fields.

Nuclear vs Atomic Binding Energy

The mass of a nucleus is less than the mass of its constituent parts due to binding energy. Binding energy is on the order of 1% the mass of the total nucleus. For an atom, masses also include the mass of the electrons minus the electronic binding energy. However, electronic binding energy is comparatively very small, less than the measurement uncertainty of the total atomic mass (note that this does not mean atomic binding energy is unknowable, it can be inferred by summing ionisation potentials for successive removal of the electrons in a given atom). Hence, we only consider nuclear binding energy, $B(Z, N)$.

$$M(Z, N) = Zm_p + Nm_n + Zm_e - B(Z, N)$$

Binding Energy per Nucleon

Binding energy alone, when plotted against A is monotonic. However, plotting binding energy per nucleon, B/A, we make the following observations.

1. The vast majority of nuclei, $20 < A < 200$ have an almost constant binding energy per nucleon, $8.3 \pm 5\%$ MeV/nucleon.
2. Maximum B/A occurs at ^{62}Ni.

3. There is a steep decline in B/A down to the lightest nuclei.
4. There is a steady and slowly steepening decline in B/A toward heavy nuclei.

We can make the following interpretations of the above observations.

1. **Constancy of B/A Beyond ^{62}Ni**: The force binding nucleons must be short ranged as it will mostly only influence its nearest neighbours, resulting in a saturation of the binding i.e. there can only be so many neighbours of a given nucleon. This is analogous to water having a constant boiling point (L_v) at STP, no matter the amount of water being heated. If the binding force were long ranged and attractive, binding energy would be non-linear and increasing.
2. **Maximum Value of B/A**: A result of the balance between the minimum surface-area-to-volume ratio and the minimum Coulomb repulsion between protons. The occurence at ^{62}Ni is coincidentally near where stellar fusion stops, but it does not mark a profound nuclear property.
3. **Steep Decline in B/A for Light Nuclei**: Also a consequence of the short range of the strong nuclear force. In light nuclei, almost all nucleons are at the surface, hence they do not have a full set of neighbours as most nucleons do in a heavy nucleus. So, the binding energy of a light nucleus is unsaturated.
4. **Slow Decline in B/A for Heavy Nuclei**: A long ranged and repulsive force would result in binding energy being non-linear and decreasing - seen in heavy nuclei because of the increasing number of protons and their long ranged Coulomb repulsion.

1.3.3 Differences in Radii and Masses

Neutron Separation Energies

By plotting differences in gross properties, shell structure becomes apparent. Figure 1.2 shows marked increase in isotope shifts at magic numbers.

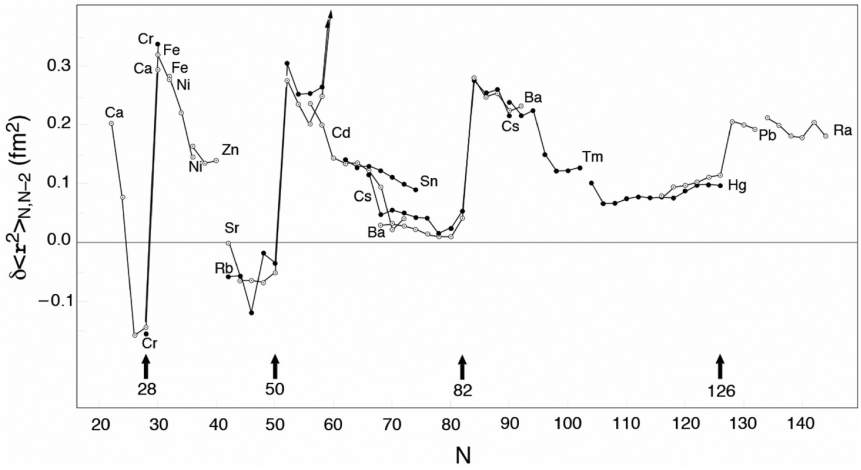


Figure 1.2: Isotope shifts for specific isotopic sequences of nuclei. Fig. 1-9 of *Primer*.

Differences in binding energies are termed separation energies. One-neutron separation energy is defined as:

$$S_n = -M(A, Z, N) + M(A-1, Z, N-1) + m_n$$

Observations to note from figure 1.4:

- Odd-even staggering of S_n.
- Discontinuities, 'steps' down, with increasing N or A.
- Between the steps, the trend is smooth and down-sloping with increasing N or A.

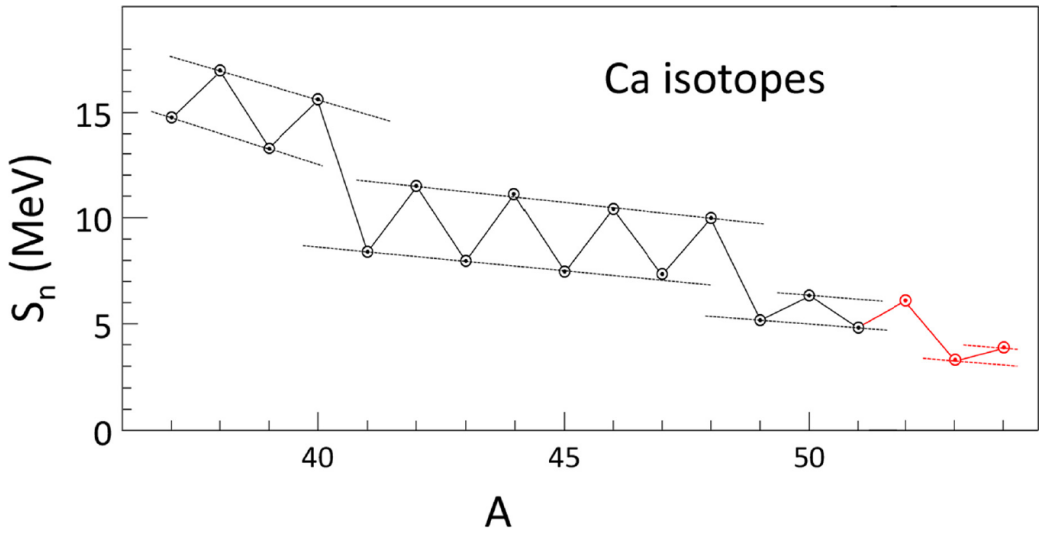


Figure 1.3: One-neutron separation energies. Data from AMDC files. Fig. 1.10 *Primer*.

We can make the following interpretations of the above observations.

- **Odd-even Staggering**: Due to enhanced binding when the neutron number is even. Similarly observed for S_p. An effect termed pairing. Manifestation of quantum mechanical correlations.
- **Steps Down**: Due to 'shell closures'. From left to right in the figure, successive neutrons are removed. Nucleons are confined in an average potential generated by all of the other nucleons. Nucleons sequentially fill orbitals. When an energy gap is reached in the removal process, there is a sudden step up in the removal energy. This effect is more clearly seen for two-neutron separation energies, figure **??**.
- **Sloping Down Between Steps**: Due to the confining potential changing with $\sim A^{1/3}$. As A increases, potential width increases and the potential becomes shallower.
- **Shell Closure Locations**: Only occur at the classical magic numbers, 20, 28, 50, 82, 126.

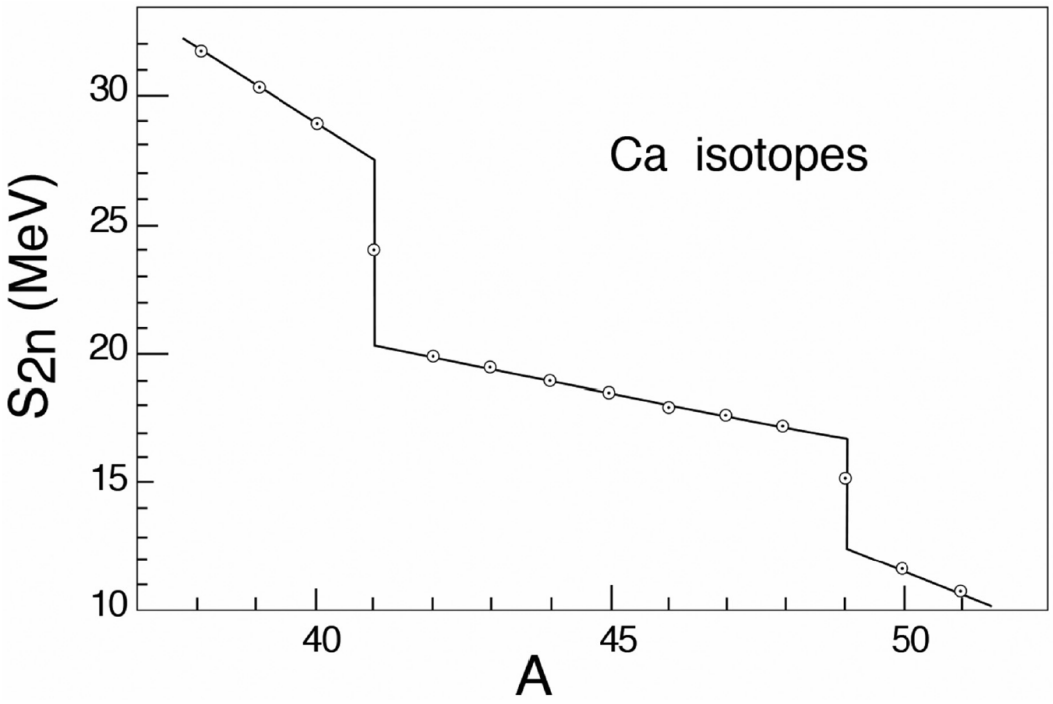


Figure 1.4: Two-neutron separation energies. Data from AMDC files. Fig. 1.11 *Primer*. The red points represent new data, measured within recent years.

Nuclear Shell Model

1.4 Prep Work for Surrey

1-4: Binding Energy Calculations

The following table shows masses and binding energies in MeV for selected nuclei. Note that these are atomic masses. Data taken from AME2016.

Isotope	$M(A,Z,N)$	$B(A,Z,N)$
n	939.57	0
^{1}H	938.78	0
^{2}H	1876.12	2.22
^{4}He	3728.40	28.30
^{5}He	4668.70	27.56
^{6}He	5606.56	29.27
^{8}Be	7456.89	56.50
^{12}C	11177.93	92.16
^{62}Ni	57685.89	545.26
^{96}Zr	89337.99	829.00
^{96}Nb	89337.83	828.38
^{96}Mo	89334.64	830.78
^{100}Zr	93073.03	852.22
^{138}Te	128480.49	1138.86
^{238}U	221742.91	1801.69

a.) How much gain in binding energy would be needed for ^{5}He to be bound in its ground state wrt disintegration into ^{4}He + n?

The difference in binding energy between the products and reactants is positive so ^{4}He is more bound than ^{5}He and energy is released

$$B(^{4}\mathrm{He}) - B(^{5}\mathrm{He}) = 28.30 - 27.56 = 0.74\,\mathrm{MeV}$$

b.) How much gain in binding energy would be needed for ^{8}Be to be bound in its ground state wrt disintegration into ^{4}He + ^{4}He?

$$2B(^{4}\mathrm{He}) - B(^{8}\mathrm{Be}) = 2(28.30) - 56.50 = 0.1\,\mathrm{MeV}$$

c.) How much energy is needed to separate a neutron from ^{6}He?

$$S_n = B(^{6}\mathrm{He}) - B(^{5}\mathrm{He}) = 29.27 - 27.56 = 1.71\,\mathrm{MeV}$$

1-6: Solar Energy Production

In the Sun, hydrogen fuses into ^{4}He. The age of the Sun is 4.6×10^9 years and it has a mass of 1.99×10^{30} kg. The energy output of the Sun is 4×10^{20} MW.

a.) Using an electron mass of 0.51MeV, how much energy is produced per atom of helium produced?

The proton-proton chain has the net effect of converting four protons into one helium-4 nucleus whilst releasing two positrons.

$$4\,^1\mathrm{H} \rightarrow {}^4\mathrm{He} + 2\mathrm{e}^+ + 2\nu_e + 2\gamma$$

The energy released is a combination of the change in binding energy and the positrons annihilating with surrounding electrons (each pair releasing twice the rest mass of the electron). First, convert the atomic mass of helium-4 to a nuclear mass: $\mathrm{m_{He}} = 3728.40 - 4(0.51) = 3726.36\,\mathrm{MeV}$.

$$4\mathrm{m}_p - \mathrm{m_{He}} - 2(2\mathrm{m}_e) = 4(938.78) - 3726.36 - 4(0.51) = 26.72\,\mathrm{MeV}$$

b.) How many kilograms of hydrogen are being fused into helium per second? Convert the energy output of the Sun to MeV.

$$\frac{4 \times 10^{26}}{1.6 \times 10^{-13}} = 2.5 \times 10^{39}\,\mathrm{MeVs}^{-1}$$

So the number of proton-proton chain reactions occuring per second is:

$$\frac{2.5 \times 10^{39}}{26.72} = 9.35 \times 10^{37}$$

So the number of hydrogen atoms fusing per second will be four times this (four ^{1}H per pp chain), 3.7×10^{38}. Converting this to moles:

$$\frac{3.7 \times 10^{38}}{N_A} = 6.22 \times 10^{14}\,\mathrm{mol}$$

Since the molar mass of hydrogen is $1\,\mathrm{gmol}^{-1}$, moles are equivalent to grams. So the total mass of hydrogen is 6.22×10^{11}kg per second.

c.) What percentage of its mass has been fused into helium so far?

Multiply the Sun's age in seconds, $4.6 \times 10^9 \times 365 \times 24 \times 3600 = 1.45 \times 10^{17}$, by the mass of hydrogen already fused, then express this as a fraction of its total mass.

$$\frac{(1.45 \times 10^{17})(6.22 \times 10^{11})}{1.99 \times 10^{30}} \times 100 = 4.5\%$$

d.) If the energy source of the Sun were chemical, how long would the Sun shine for the given energy output? Assume an energy yield of 10 eV per hydrogen atom.

1.99×10^{30} kg of hydrogen corresponds to 1.99×10^{33} mol or 1.2×10^{57} atoms of hydrogen. Each of these atoms contributes 10 eV, so a total energy of 1.92×10^{39} J. So for an energy output of 4×10^{20} MW, the Sun will last for:

$$t = \frac{1.92 \times 10^{39}}{4 \times 10^{26}} = 4.8 \times 10^{12}\,\text{s} = 1.5 \times 10^{5}\,\text{years}$$

1-7: Fission of Uranium

Uranium can undergo fission, for example ^{238}U $\rightarrow$ ^{138}Te + ^{100}Zr. How much energy is released per kilogram?

$$\begin{aligned}\Delta B &= B(^{138}\text{Te}) + B(^{100}\text{Zr}) - B(^{238}\text{U}) \\ &= 1138.86 + 852.22 - 1801.69 \\ &= 189.39\,\text{MeV per fission}\end{aligned}$$

In 1kg of ^{238}U there are of the order $\frac{1000\text{g}}{238\text{gmol}^{-1}} \times N_A \approx 2.53 \times 10^{24}$ nuclei of ^{238}U. So the energy released per kg is

$$189.39 \times 2.53 \times 10^{24} = 4.79 \times 10^{26}\,\text{MeV kg}^{-1}$$

1-11: Electronic and Nuclear Binding Energy for Deuterium

For an atom of ^{2}H, what is the ratio of electronic binding energy to nuclear binding energy?

The atomic ground state of ^{2}H is approximately 13.6 eV. Comparing this to the nuclear binding energy, 2.22 MeV:

$$\frac{13.6}{2.22 \times 10^6} \approx 6 \times 10^{-6} = 0.0006\%$$

1-13: Finding the Line of Stability in Mass Parabolas

The Bethe-Weizsäcker formula for the binding energy of nuclei is expressed as

$$B(A,Z,N) = a_1 A + a_2 A^{2/3} + a_3 \frac{Z^2}{A^{1/3}} + a_4 \frac{(N-Z)^2}{A}$$

where the approximate values of the coefficients in MeV are $a_1 = 15.68$, $a_2 = -18.56$, $a_3 = -0.717$ and $a_4 = -28.1$. The four terms are called the volume, surface, Coulomb and symmetry terms respectively. Note that there is no pairing term. Using

$$M(Z,N) = Zm_p + Nm_n + Zm_e - B(Z,N)$$

what would be the value of Z for the most stable isotope with:
a.) $A = 133$? Compare with figure ??
b.) $A = 134$? Compare with figure ??
c.) $A = 294$?
d.) $A = 400$?
Hint: Maximise B versus Z at constant A and substitute $N = A - Z$ before starting
Begin by casting B in terms of two variables. Substitute $N = A - Z$.

$$B(A,Z) = a_1 A + a_2 A^{2/3} + a_3 \frac{Z^2}{A^{1/3}} + a_4 \frac{(A-2Z)^2}{A}$$

Then differentiate wrt Z at constant A, set to zero to minimise.

$$\frac{\partial B}{\partial Z} = 2a_3 \frac{Z}{A^{1/3}} - 4a_4 \frac{(A-2Z)}{A} = 0$$

Rearrange to get Z_{min} in terms of A.

$$Z_{min}(A) = \frac{2a_4A}{a_3A^{2/3} + 4a_4} = \frac{A}{\frac{a_3}{2a_4}A^{2/3} + 2}$$

Which gives
a.) $Z_{min}(133) = 57.0$ MeV
b.) $Z_{min}(134) = 57.4$ MeV
c.) $Z_{min}(294) = 114.7$ MeV
d.) $Z_{min}(400) = 148.6$ MeV

1-14: Investigating the Bethe-Weizsäcker Formula

Explore the contribution of the four terms in the Bethe-Weizsäcker binding energy formula by setting, in turn, the parameters a_1 to a_4 to zero. How would such changes compress or expand the reach of the table of isotopes? Explore particularly the effects on the stability of the actinide isotopes and the relative stability of even-even, odd-A and odd-odd isotopes.

Setting $a_1 = 0$ removes the volume term, which represents the SNF between nearest neighbours. It is the only positive term in the SEMF, so removing it causes the binding energy to be negative. So, no bound nuclei exist, including actinides. All nuclei would separate into their constituent nucleons.

Setting $a_2 = 0$ removes the surface term, it represents the reduction in binding energy due to nucleons at the surface (the ones with fewer nearest neighbours than those in the interior). It is more apparent for light nuclei, removing it causes an increase in binding energy, since nucleons at the surface would contribute the same amount to the total binding energy as those in the interior. This means that light nuclei would be more bound and so the stability valley widens, since the drip lines are further apart. For actinides, the surface term is the main barrier preventing heavy nuclei from spontaneously undergoing fission. So the chart of stability truncates for heavy elements (they would all undergo spontaneous fission without any activation barrier).

Setting $a_3 = 0$ removes the Coulomb term, it represents the long range electrostatic repulsion between protons. It causes heavy nuclei to because

unstable and so pushes the line of stability towards $N > Z$. Removing it causes the line of stability to become $Z = N$ for all A. Actinides become stable against alpha decay and spontaneous fission and their most stable forms would become symmetric in N and Z. Also, without a build-up of charge, nuclei would be able to grow arbitrarily.

Setting $a_4 = 0$ removes the symmetry term, it represents the tendency for nuclei to balance the number of protons and neutrons. Removing it causes an increase in binding energy of asymmetric nuclei and for the Coulomb term to provide the only Z dependence. Since protons repel each other, the $Z > N$ side of the line of stability collapses and the $N > Z$ side expands. So, proton-rich actinides become more unstable and readily undergo beta-plus decay and heavy elements would only exist as almost pure neutron-rich matter.

Note on relative stability of even-even, odd-A and odd-odd isotopes: The stability is governed by the parity term $\delta(A, Z)$ which increases the binding energy for even-even, decreases for odd-odd and adds nothing for odd-A. So for odd-A mass chains the first four terms entirely govern the stability. For even-A mass chains, the pairing term splits the A chain into two parabolas separated by 2δ. The Coulomb and asymmetry terms determine the steepness of the parabola.

1.5 Nuclear Excitation Patterns

2-5: Exploring γ Transition Energy Ratios

The ratios r_6 and r_8 are defined for nuclei which approximate an ideal rotor as

$$r_6 = \frac{E(6_1^+) - E(4_1^+)}{E(2_1^+)} \qquad r_4 = \frac{E(4_1^+) - E(2_1^+)}{E(2_1^+)}$$

a.) Extend this study by defining r_8

$$r_8 = \frac{E(8_1^+) - E(6_1^+)}{E(2_1^+)}$$

b.) Obtain the rotor limit value for r_8

Using $E(I) \propto I(I+1)$ where I is the total angular momentum quantum number.

$$r_8 = \frac{8(8+1) - 6(6+1)}{2(2+1)} = 5$$

c.) Using ENSDF for even-even rare earth and actinide nuclei around ^{172}Yb and ^{242}Pu, make a plot of r_8 vs r_2

2-6: Nuclear Recoil Correction for Gamma Transitions

To obtain high precision on ratios of level energies as shown in figure ?? it may be necessary to perform a correction to the gamma-ray energies observed in experiment as gamma-ray emission is a two-body process where the emitting nucleus recoils. So some small part of the transition energy is taken up by the recoil.
a.) Given that the photon's energy and momentum are related by $E = pc$, make use of energy and momentum conservation to derive a formula for the nuclear recoil energy as a function of gamma-ray energy, E_γ and atomic mass, A of the recoiling nucleus.

By conservation of momentum

$$p_\gamma = p_{\text{nuc}}$$
$$\frac{E_\gamma}{c} = p_{\text{nuc}}$$

So then the recoil energy of the nucleus will be

$$E_{\text{recoil}} = \frac{p_{\text{nuc}}^2}{2m_{\text{nuc}}} = \frac{E_\gamma^2}{2Ac^2}$$

where A is in amu. So for the units to work out, use $c^2 = 931.5 \times 10^6$ eV/u.

b.) Estimate the magnitude of the recoil correction for $A = 240$ and $E_\gamma = 150$ keV.

$$E_{\text{recoil}} = \frac{(150 \times 10^3)^2}{2 \times 240 \times 931.5 \times 10^6} = 0.05\,\text{eV}$$

c.) Estimate the magnitude of the recoil correction for $A = 12$ and $E_\gamma = 3213.79$ keV, $E_\gamma = 4438.94$ keV. Note that these are the energies of γ rays de-exciting the Hoyle state in ^{12}C. Thus, give your estimate of the excitation energy of the Hoyle state and compare with the value given in ENSDF .

For $E_\gamma = 3213.79$ keV:

$$E_{\text{recoil}} = \frac{(3213.79 \times 10^3)^2}{2 \times 12 \times 931.5 \times 10^6} = 462.00\,\text{eV}$$

For $E_\gamma = 4438.94$ keV:

$$E_{\text{recoil}} = \frac{(4438.94 \times 10^3)^2}{2 \times 12 \times 931.5 \times 10^6} = 881.38\,\text{eV}$$

So the total excitation energy of the Hoyle state is

$$3213.79 + 4438.94 + 0.462 + 0.88138 = 7654.07338\,\text{keV}$$

This matches the adopted level from the ENSDF, excerpt below.

E(level)	J^π	$T_{1/2}$	XREF	Comments
7654.07 *19*	0^+	9.3 eV *9*	AB DE GHIJK MNOP TUVWX	XREF: Others: AA, AC, AD, AE, AF, AI, AJ, AK, AL, AN, AO, AP, AQ, AR, AU, AV, AW, AX, AY, AZ, BA, BB, BC, BD, BG, BH, BI %IT=4.16×10^{-2}; %α≈100 T=0 E(level): See discussion in (1976No02). Note: E_x=7657.8 keV *10* is obtained from analysis of γ rays measured in (2016Mu06).

2-11: Exploring the Nuclear Structure of ^{152}Sm with ENSDF Pages

Part 1: Levels

Looking at adopted levels and gammas. First lot of tables gives the levels. Second lot gives the gammas that connect the levels.

First table gives energy, spin and parity, half life, flags (what reactions it was produced in) and comments. Easier to compute separation energies from the AMDC

If spin and parity, J^{π} is in brackets then its not firmly established. More XREF flags give more confidence. Footnotes contain info on how the data was manipulated but also assignments to different rotational bands.

f are conversion coefficients, calculated with the bricc tool.

Part 2: Matching Data to Levels

22MeV deuteron scattering experiment. Unpublished so doesn't feature in the ensdf. Magnetic spectrograph used to analyse the scattered deuterons at a fixed angle of 60degrees with a high momentum bite setting. The data has FWHM energy resolution of about 3keV (much lower fidelity than possible by measuring gamma ray transitions with e.g. a high purity germanium detector). Consequently, the 1085 peak is seen as a single peak but is actually two closely spaced levels that aren't resolved

As to how data is compiled in ENSDF, the energies of excited states are adopted from the highest precision measurements that are available. Usually γ ray transition measurements. For multiple sources, a weighted average of the most precise sources is adopted. To adopt a given excited state energy, a whole network of γ transition measurements is used, both de-exciting and populating a given level. Then, a least-squares fit is taken over the whole network. For γ ray energies with a very high precision (down to eV), then the recoil energy of the nucleus must be added to arrive at the transition energy.

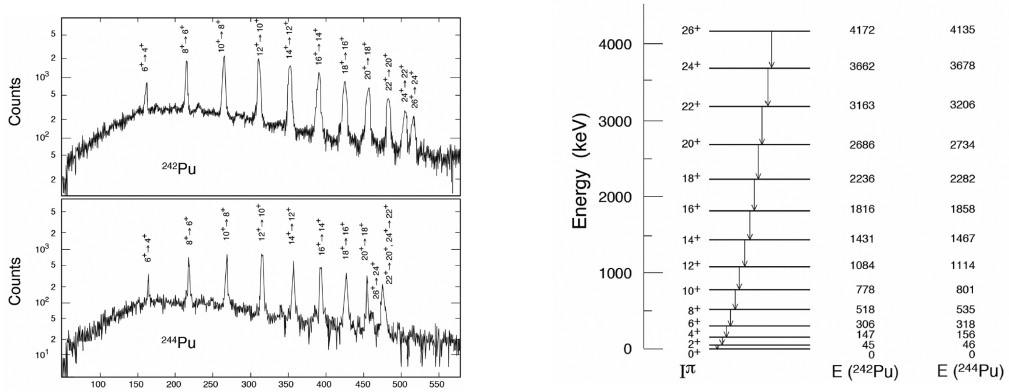
Part 3: Matching Data to Transitions Connecting High Spin States

Here we consider two data sets - Coulomb excitation (coulex) and the nuclear reaction ^{150}Nd$(\alpha, 2\,\mathrm{n}\gamma)^{152}$Sm. I_γ is the relative intensity, where 100 is the strongest transition observed. The internal conversion coefficients, α, are calculated using BrIcc, not measured. They are larger for low energy transitions and for high Z nuclei. Yrast states are the lowest energy states of a given angular momentum, forming the ground state rotational band.

1.6 Nuclear Deformation and Rotations

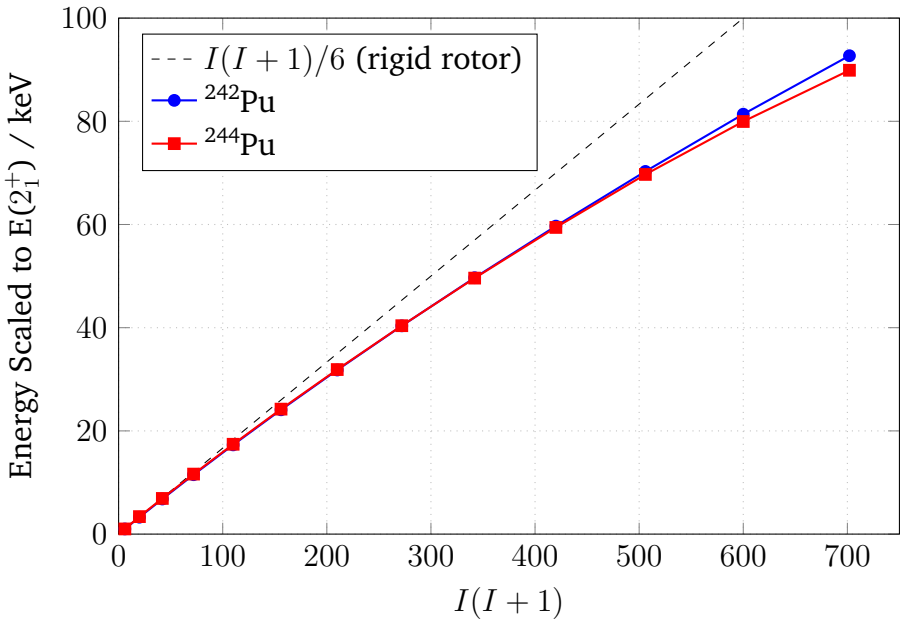
3-1: Rotational Bands in Plutonium Isotopes

Below are the ground state rotational bands in ^{242}Pu and ^{244}Pu along with the excitation energies of the rotational states.



Scale the excitation energies to that of 2_1^+ states i.e. divide the energies for ^{242}Pu by 45 and ^{244}Pu by 46. Having scaled the energies, plot the two bands as $E(I_{\text{scaled}})$ vs $I(I+1)$. Note that, while the energies look significantly different, when scaled they are nearly identical i.e. there is evidently some underlying universal rotational energy profile. This does not appear to have been recognised, but holds for the most strongly deformed ground-sate bands, extending into the rare earth region e.g. ^{174}Yb. Higher precision energy values can be used but this exercise captures the essence of the feature.

I	$I(I+1)$	E(^{242}Pu)	E_{scaled}(^{242}Pu)	E(^{244}Pu)	E_{scaled}(^{242}Pu)
2	6	45	1.00	46	1.00
4	20	147	3.27	156	3.39
6	42	306	6.80	318	6.91
8	72	518	11.51	535	11.63
10	110	778	17.29	801	17.41
12	156	1084	24.09	1114	24.22
14	210	1431	31.80	1467	31.89
16	272	1816	40.36	1858	40.39
18	342	2236	49.69	2282	49.61
20	420	2686	59.69	2734	59.43
22	506	3163	70.29	3206	69.70
24	600	3662	81.38	3678	79.96
26	702	4172	92.71	4135	89.89



3-6: Exploring the Nuclear Structure of ^{175}Lu with ENSDF Pages

1.7 Towards a Global View of Nuclear Structure

4-4: J and M Coupling

Using the m scheme, deduce the possible couplings in terms of J and M for:

a.) Two identical fermions with $j = 7/2$ e.g. coupling two $f_{7/2}$ neutrons. Each particle will have eight possible states (projections), $\pm\frac{1}{2}, \pm\frac{3}{2}, \pm\frac{5}{2}, \pm\frac{7}{2}$. Since the particles are identical, they cannot occupy the same m states so $m_1 \neq m_2$

b.) Three identical fermions with $j = 7/2$ and obtained the allowed spins for the $v = 3$ states. Compare your result with information in figure ??.

c.) Two non-identical fermions with $j = 7/2$ e.g. coupling an $f_{7/2}$ proton and $f_{7/2}$ neutron.

4-10: Mirror Nuclei on the $A = 42$ Isobar

Make a comparison of the observed energies in ENSDF for the states in ^{42}Ca, ^{42}Sc and ^{42}Ti below 1600 keV as a side-by-side plot. Noting your answers to 4-4a, and that fact that $^{42}_{20}$Ca and $^{42}_{22}$Ti are a mirror pair, what can you conclude about the behaviour of the two valence nucleons in these $A = 42$ isobars?

4-11: Exploring the Structure of ^{207}Pb and ^{208}Pb with ENSDF Pages

The following are notes from video exercises exploring different aspects of the structure of ^{207}Pb intended to encourage familiarity with identifying states adjoining the doubly magic nucleus, ^{208}Pb.

4-6: Classification of the Lowest States in ^{209}Bi

Using ENSDF and information from the figures below, classify the lowest states in ^{209}Bi i.e. shell-model particle states, shell-model hole states and core-excited states coupled to shell-model particle states.

1.8 Simple Models of Nuclear Structure

5-4:

Illustrate the difference in rotational energies for nuclei of mass A and $A + 1$ as depicted in plots of E_γ versus I_i. Use the classical expression for the moment of inertia of a sphere.

Moment of inertia of a sphere is given by:

$$I = \frac{2}{5} M R^2$$

1.9 Nuclear Spectroscopic Measurements

6-1: Radii in the Bohr Model

The Bohr radius, a_0 is the radius of an electron in the ground state of a Hydrogen atom

$$a_0 = \frac{4\pi\epsilon_0\hbar^2}{m_e e^2}$$

The corresponding radius for the first Bohr orbit for an element with higher Z is

$$r_Z = \frac{a_0}{Z}$$

a.) What would be the radius of the first Bohr orbit for a muon bound to $^{150}_{60}$Nd?

For a muonic atom, replace $m_e \rightarrow m_\mu$ to give $a_0 = 2.55 \times 10^{-13}$. Then substitute as follows.

$$r_{60} = \frac{2.55 \times 10^{-13}}{60} = 4.25\,\text{fm}$$

b.) Is this orbit, on average, 'inside' the nucleus?

Assuming a spherical nucleus, $R = R_0 A^{1/3} = 1.05 \times 10^{-15}(150)^{1/3} = 5.57\,\text{fm}$. Since $4.25 < 5.57\,\text{fm}$, yes this orbit is 'inside' the nucleus.

c.) Using the concept of the first and second Bohr orbits, can the energy scale in the following spectra of muonic atom transitions be simply explained?

6-4: Precision of a Penning Trap

The precision on nuclear masses which can be obtained depends on the number of ions in the measurement, N, and the mass resolving power, R:

$$\frac{\delta m}{m} \propto \frac{1}{R\sqrt{N}}$$

The mass resolving power for a Penning trap is typically 10^5 for an observation time of 100 ms. This means that the isotope under study must be sufficiently long-lived to allow a measurement to be conducted for 100 ms. Estimate the number of ions that would need to be trapped to achieve a mass precision, $\delta m/m = 10^{-7}$

Rearrange the above equation for N.

$$N = \frac{1}{\left(R\frac{\delta m}{m}\right)^2} = \frac{1}{(10^5 \times 10^{-7})^2} = 10^4 \text{ ions}$$

6-5: Lutetium-176

The isotope ^{176}Lu occurs naturally. Explain this in terms of expectations based on masses of $A = 176$ isobars and spins of states in ^{176}Lu and its neighbours.

6-6: Isotopes of Uranium

The isotopes ^{235}U and ^{238}U both occur naturally. They possess relative natural abundances of 0.7204% and 99.2742%, and half lives of 7.04×10^8 y and 4.468×10^9 y respectively. Assuming that these two isotopes were present in equal amounts when they were formed (in some coalescing binary neutron star pair) estimate how long ago that was. Compare this with the estimated age of the Earth, 4.5×10^9 y.

6-7: The Fraction of Plutonium in Uranium Ore

The longest lived isotope of plutonium is ^{244}Pu with a half life of 8.13×10^{7} y. Assuming that plutonium and uranium have similar mineral chemistry and were formed in equal amounts as in the scenario in 6-6, estimate the ratio ^{244}Pu/^{238}U in uranium ore. Has this been investigated?

Questions for David What is the difference between an evaluation and a compilation of nuclear data

How do you choose the reference point for an isotope shift? In the Bethe-Weizsacker formula (1-13) why is there no pairing term? Removing the surface term - would there be much change for heavy nuclei? their B would increase, but comparatively not as much as small ones.

square vs curved brackets in the ensdf

Part II

Taught Programme: Instrumentation

Part III

Taught Programme: Simulation and Modelling

Part IV

Taught Programme: Nuclear Data Pipeline

Part V

Taught Programme: Consolidation

Part VI

Project